PRE-ALGEBRA (7TH) · UNIT 10: AREA AND VOLUME

Backyard Pool Makeover

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design a backyard with a pool, patio, garden, and fire pit. Area formulas for parallelograms, triangles, trapezoids, and circles price the hardscape; surface area prices the pool liner; and volume finds the water.

Comparing a rectangular and a round pool shows how shape affects area, volume, and cost.

Standards Addressed

- 7.G.B.4 Area and circumference of circles
- 7.G.B.6 Area, volume, and surface area of 2D and 3D figures
- 8.G.C.9 Volume of cylinders
- Honors: scale factors for area and volume

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with backyard photos. Part 1 patio and garden areas.
Day 2	Part 2 pool volume and surface area.
Day 3	Part 3 budget and proposal.

Materials

- Student handout and Price Cards
- Grid paper and rulers
- Compasses
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Single-shape areas, two pool volumes, gallons, and a simple budget.
CP1	Composite patio, mulch volume in cubic yards, liner surface area, water cost, and a budget comparison.
Honors	Rectangle-plus-semicircle patio, ring volume, sloped pool, surface-area-to-volume, and optimizing the radius.

Watch For (Common Misconceptions)

- Using diameter instead of radius.
- Forgetting to convert cubic feet to cubic yards or gallons.
- Including the top of the pool in the liner area.

Extension Ideas

- Price real pool supplies.
- Design a hot tub (cylinder) and compare.
- Build a scale model with cardboard.

Answer Key – CP2

Part 1: Patio & Garden

#	Task	Answer
1	The patio is a parallelogram with base 20 ft and height 12 ft. Find its area.	20 × 12 = 240 sq ft
2	Find the cost of patio stone for the patio.	240 × $12 = $2,880
3	A triangle garden has base 10 ft and height 8 ft. Find its area.	$\frac{1}{2}(10)(8) = 40$ sq ft
4	A trapezoid flower bed has bases 6 ft and 10 ft and height 4 ft. Find its area.	$\frac{1}{2}(6 + 10)(4) = 32$ sq ft

Part 2: The Pool

#	Task	Answer
5	Find the floor area of the rectangular pool (30 ft × 15 ft).	450 sq ft
6	Find the floor area of the round pool (diameter 24 ft).	r = 12; 3.14 × 144 ≈ 452.16 sq ft
7	Find the volume of water in the rectangular pool (5 ft deep).	30 × 15 × 5 = 2,250 cu ft
8	Find the volume of the round pool (4 ft deep).	452.16 × 4 ≈ 1,808.64 cu ft

Part 3: Fire Pit & Cost

#	Task	Answer
9	Find the area of the fire pit.	3.14 × 4 = 12.56 sq ft
10	How many gallons of water fill the rectangular pool? (1 cu ft ≈ 7.5 gal)	2,250 × 7.5 = 16,875 gallons
11	Choose a pool and find the cost of its floor. Add the patio. Is it under $25,000?	Rectangular: 450 × 45 = $20,250 + $2,880 = $23,130 (yes). Round: 452.16 × 40 ≈ $18,086 + $2,880 = $20,966 (yes).

Reflection

#	Prompt	Look for
R1	Why is pool size measured with volume but the patio with area?	The pool holds water in three dimensions; the patio covers a flat surface.
R2	How did you use the radius in the circle formulas?	Area uses r^2; the diameter must be halved first.

Answer Key – CP1

Part 1: Hardscape

#	Task	Answer
1	The patio is a trapezoid with parallel sides 26 ft and 18 ft and height 14 ft. Find its area and cost.	$\frac{1}{2}(26 + 18)(14) = 308$ sq ft; \$3,696
2	A circular fire pit (r = 2 ft) sits in the patio and is not paved. Find the paved area and the new cost.	308 − 12.56 = 295.44 sq ft; \$3,545.28
3	Mulch 3 inches deep covers a 10 ft by 8 ft triangle garden. How many cubic yards? (27 cu ft = 1 cu yd)	Area 40 sq ft × 0.25 ft = 10 cu ft ≈ 0.37 cu yd
4	What percent of the yard is the patio?	308 ÷ 2,400 ≈ 12.8%

Part 2: Pools

#	Task	Answer
5	Find the volume in cubic feet and gallons of each pool.	Rect: 2,250 cu ft = 16,875 gal. Round: π(144)(4) ≈ 1,808.64 cu ft ≈ 13,565 gal
6	Find the liner area for each pool (floor + walls).	Rect: 450 + 2(30·5) + 2(15·5) = 900 sq ft. Round: 452.16 + 2π(12)(4) ≈ 452.16 + 301.44 = 753.6 sq ft
7	Find the liner cost and the cost to fill each pool with water.	Rect: \$2,250 liner, \$101.25 water. Round: \$1,884 liner, about \$81.39 water.
8	Which pool holds more water per square foot of yard it covers?	Rect: 2,250 ÷ 450 = 5 cu ft per sq ft; round: 4. The rectangular pool.

Part 3: Budget Proposal

#	Task	Answer
9	Find the total cost for each pool option (floor, liner, water) plus the patio.	Rect: 20,250 + 2,250 + 101.25 + 3,545.28 = \$26,146.53. Round: 18,086.40 + 1,884 + 81.39 + 3,545.28 = \$23,597.07
10	Which option fits the budget? How much is left for mulch and the garden?	Round pool; \$1,402.93 left
11	Write your proposal with a labeled yard drawing and a cost table.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why is the liner area different from the floor area?	The liner covers the floor and the walls, so it is the surface area without the top.

#	Prompt	Look for
R2	How did changing the shape of the pool change the cost?	Different shapes have different floor and wall areas and volumes, which changed every cost.

Answer Key – Honors

Part 1: Patio Design

#	Task	Answer
1	The patio is a 20 ft by 16 ft rectangle with a semicircle (diameter 16 ft) on one end. Find its area and cost of stone.	$320 + \frac{1}{2}\pi(64) = 320 + 32\pi \approx 420.53$ sq ft; ≈ $5,046.37
2	Find the perimeter of the patio for edging.	The semicircle replaces one 16-ft side: straight sides 20 + 16 + 20 = 56 ft plus half the circumference 8π ft → 56 + 8π ≈ 81.13 ft
3	A fire pit is a cylinder ring: outer radius 2.5 ft, inner radius 2 ft, height 1.5 ft. Find the volume of stone in the ring.	$\pi(6.25 - 4)(1.5) = 3.375\pi \approx 10.60$ cu ft
4	How many cubic yards of concrete is that?	10.60 ÷ 27 ≈ 0.39 cu yd

Part 2: Pool Engineering

#	Task	Answer
5	A sloped rectangular pool is 3 ft deep at one end and 8 ft at the other (length 30 ft, width 15 ft). Its side is a trapezoid. Find the volume.	Trapezoid side area $\frac{1}{2}(3 + 8)(30) = 165$ sq ft × 15 = 2,475 cu ft
6	Find the ratio of liner surface area to volume for the round pool and the flat rectangular pool. Which is more efficient?	Round: 753.6 ÷ 1,808.64 ≈ 0.42. Rect: 900 ÷ 2,250 = 0.40. The rectangular pool uses slightly less liner per cubic foot.
7	What diameter would a 4-ft-deep round pool need to hold the same volume as the flat rectangular pool?	$\pi r^2(4) = 2{,}250 \rightarrow r^2 \approx 179.05 \rightarrow r \approx 13.4$ ft → d ≈ 26.8 ft
8	If every dimension of the round pool were doubled, how would its volume and liner area change?	Volume × 8, liner area × 4

Part 3: Optimize

#	Task	Answer
9	Write the total cost of a round pool with radius r (4 ft deep) including floor ($40/sq ft) and liner ($2.50/sq ft), as an expression in r.	$40\pi r^2 + 2.5(\pi r^2 + 8\pi r) = 42.5\pi r^2 + 20\pi r$
10	With a $5,046 patio, find the largest whole-foot radius that keeps the total under $25,000.	$42.5\pi r^2 + 20\pi r \le 19{,}954 \rightarrow$ r = 11: 16,155 + 691 = 16,846 ✓; r = 12: 19,227 + 754 = 19,981 ✗ → r = 11 ft
11	Present your final design and justify each choice with a calculation.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does doubling every dimension multiply volume by 8?	Volume multiplies three lengths: $2 \times 2 \times 2 = 8$.
R2	What trade-offs did you face between pool size and budget?	Larger radius increases floor cost quadratically, using up the budget quickly.